MILITARY SPECIFICATION

CARTRIDGE, 81MM, TP, M362A1
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of target practice cartridge designated as M362A1, for use with the 81MM Mortar.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-G-2550 - General Specification for Ammunition Except Small Arms Ammunition.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

8839446 - Cartridge, TP, M362A1 W/Fuse M524 (T186E11) for 81MM Mortar Assembly.
7549031 - Container, Ammunition, Fiber, M252, for 81MM Mortar Ammunition. Assembly, Details and Packing.
7549032 - Box, Packing, Ammunition, for 81MM Mortar Ammunition and Fiber Container M252. Assembly, Details and Packing.

FSC 1315

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PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-(see 6.3) - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Cartridges.-The cartridge shall comply with all requirements specified on Drawing (dwg.) 8839446, and with all requirements specified in applicable specifications.

3.3 Proving ground test.-The cartridge shall comply with the following requirements when tested as specified in 4.4.1.

3.3.1 Failure to fire.-The cartridge shall not fail to fire.

3.3.2 Premature bursts.-There shall be no premature burst.

3.3.3 Impact functioning.-The cartridge shall function upon impact.

3.3.4 Range.-No cartridge shall have a horizontal range less than 80 percent of the average horizontal range.

3.4 Workmanship.

3.4.1 All parts shall be free of chips, dirt, grease, rust and other foreign matter. The cleaning method shall not be injurious to the parts nor shall the parts be contaminated by the cleaning agents.

3.4.2 The painting and markings shall be restored by the loading contractor on any components furnished to him painted and marked when such painting and marking have been marred or obliterated.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier

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may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies procured under this specification and shall perform all inspections required by the system. A current written description of the system shall be submitted to the contracting officer, and meet his approval, prior to initiation of production. It shall include a description covering the controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (gages, measuring, and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.3 and 4.4 (except proving ground tests), and the other documents referenced herein. The contractor shall not be restricted to the inspection station nor the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that affects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

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b. Number of units of product inspected.
c. Results obtained for all inspection performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of analyses on all material procured directly by the contractor.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification shall consist of: (a) surveillance of contractor operations to determine that the methods and procedures of the written procedure are being properly applied and (b) Government product inspection. The Government inspector shall also ascertain prior to acceptance that the product complies with all the quality assurance provisions of other specifications referenced in any of the contractual documents.

4.1.3.1 Surveillance.-Surveillance shall include, but is not limited to:

a. Observation of procedures concerning lot formation and identification.
b. Observation of sampling procedures and application of acceptance criteria.
c. Determination that all required examinations and tests are performed in accordance with the prescribed procedures of this specification, or approved equivalents thereto.
d. Review of procedures for control and disposition of nonconforming material.

4.1.3.2 Product inspection.-Physical inspection by the Government of contractor accepted product will be performed to the extent necessary to assure contractor compliance with the requirements of this specification and to determine the reliability of contractor records. Severity of Government inspection of individual characteristics will be directly related to the seriousness classification assigned. In no instance will a characteristic classified "critical"

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be accepted solely on the basis of the contractor's records. Specification MIL-G-2550 will apply.

4.1.3.3 Deviations and deficiencies.-Deviations from the prescribed or agreed upon procedures, or instances of poor practice which might have an effect on the quality of the product, shall be called to the attention of the contractor immediately. To avoid interference, with operations, the contractor shall designate a responsible official, or officials, to whom the Government inspector will report such instances. Failure of the contractor to correct deficiencies promptly shall be cause for suspension of acceptance until correction has been made or until conformance to prescribed criteria has been demonstrated.

4.2 Preproduction inspection.

4.2.1 Submission.-The contractor shall submit 40 consecutively produced cartridges, which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government-approved facility designated by the contracting officer. The preproduction quantity shall be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, or specification, such as to significantly affect product uniformity as determined by the Government inspector.

4.2.2 Inspections to be performed.-Each assembly shall be inspected by the inspector designated by the procurement activity, for all the complete round (end item) requirements specified on the applicable drawing and in this specification. Upon completion of these inspections the 40 cartridges shall be forwarded to a Government proving ground for ballistic testing in accordance with 4.4.1. The contracting officer shall furnish instructions for shipment of the ballistic sample.

4.2.3 Rejection.-If any assembly fails to comply with the applicable requirements, the preproduction quantity shall be rejected and the contractor shall take the necessary corrective action and submit a new preproduction quantity, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary, until such time as a quantity successfully passes the inspection specified or until otherwise directed by the procuring activity.

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4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of cartridges produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance, or fit, as determined by the Government, shall not necessitate changing the lot number. Each lot shall contain:

a. Metal parts or metal parts assemblies of one interfix lot number from one manufacturer.

b. Loaded ignition cartridges of one interfix lot number from one manufacturer.

c. Fin assemblies of one interfix lot number from one manufacturer.

d. Loaded primers of one interfix lot number from one manufacturer.

e. Propellant increments made from not more than one lot of propellant from one manufacturer.

f. Loaded fuzes of one interfix lot number from one manufacturer.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Shell loading assembly, prior to loading (see dwg. 8839447 covering a detail of dwg. 8839446).

Categories	Defects	Method of inspection
Critical:	None defined	
Major:	None defined	
Minor:	AQL 1.0 percent	
201.	Foreign matter in shell cavity (see 3.4) ..	Visual
202.	Interior protective coating with bare spot	Visual

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4.3.2.2 Shell loading assembly (see dwg. 8839447 covering a detail of dwg. 8839446).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
101.	Weight improper	Balance
Minor:	AQL 2.5 percent	
201.	Total length, maximum (max.)	Gage
202.	Depth of fuze cavity, max.	Gage
203.	Protective coating with bare spot	Visual
204.	Marking misleading or unidentifiable	Visual
205.	Evidence of poor workmanship (see 3.4)	Visual

4.3.2.3 Assembly (see dwg. 8839446).

Categories	Defects	Method of inspection
Critical:		
1.	Primer above flush with rear face of fin relief	Straight edge
2.	Fuze safety wire missing or improperly secured	Visual-Manual
3.	Ignition cartridge missing or inverted	Visual
4.	Warning label missing or illegible	Visual
Major:	AQL 0.40 percent	
101.	Fin assembly loose	Visual-Manual
102.	Any component damaged or incorrectly assembled	Visual
103.	Incorrect number of propellant increments	Visual
Minor:	AQL 1.5 percent	
201.	Fuze loose	Visual-Manual
202.	Exterior protective coating with bare spot, except for slight scratches	Visual
203.	Marking misleading or unidentifiable	Visual
204.	Evidence of poor workmanship (see 3.4)	Visual

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4.3.2.4 Unsealed fiber container (see dwg. 7549031).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.25 percent	
101.	Packing stop missing	Visual
102.	Cartridge support missing	Visual
Minor:	AQL 1.5 percent	
201.	Glue or asphalt on round	Visual
202.	Cartridge improperly assembled into container	Visual
203.	Round cannot be removed by hand	Manual

4.3.2.5 Sealed fiber container (see dwg. 7549031).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Container damaged or cut through all asphalt layers	Visual
102.	Inner layer of tape not in contact with container for 1/2 inch on each side of joint	Visual
103.	Metal end loose or distorted	Visual-Manual
Minor:	AQL 2.5 percent	
201.	Gap between cover and body of container exceeding 1/8 inch	Visual
202.	Tear tab length inadequate	Visual
203.	Cuts, scuffs, or gouges in outer layer ..	Visual
204.	Marking misleading or unidentifiable	Visual
205.	Contents loose	Manual

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4.3.2.6 Sealed packing box (see dwg. 7549032).

Categories	Defects	Method of inspection
Critical:	None defined.	
Major:	AQL 0.40 percent	
101.	Box damaged exposing contents	Visual
102.	DOD symbol missing or incorrect	Visual
103.	Board broken or split	Visual
104.	Hardware or strapping missing, broken or loose	Visual-Manual
Minor:	AQL 1.5 percent	
201.	Marking misleading or unidentifiable	Visual
202.	Car seal missing, unsealed or improperly positioned	Visual
203.	Handle missing or insecure	Visual-Manual
204.	Hardware or strapping improperly engaged or mislocated	Visual-Manual

4.3.3 Testing.-Testing equipment shall be in accordance with 4.3.4.

4.3.3.1 Check test for possible deterioration of primers (see applicable primer specification).-If the total elapsed time between original acceptance of any primer lot and the assembly of that lot into complete round cartridges exceeds two years, or if the primers have been exposed to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass, the check test specified in applicable loading specifications immediately before the primer lot is assembled into the cartridge. The test shall be performed by the contractor on primers selected by the Government inspector at the facility assembling the complete cartridge (see 6.2).

4.3.3.2 Proving ground test; failure to fire (see 3.3.1), impact functioning (see 3.3.3) and range (see 3.3.4) Major defect; and premature bursts (see 3.3.2) - Critical defect.-A sample of 40 cartridges shall be selected from each lot for this test. If a premature burst occurs, or three or more rounds fail to comply with the requirements of 3.3.1, 3.3.3 and 3.3.4, the lot shall be rejected. If no premature burst occurs, or if only one or two rounds fail to comply with the requirements of 3.3.1, 3.3.3 and 3.3.4, a second sample of 80 assemblies shall be tested. If a premature burst occurs during retest, or if the combined number of firing impact functioning and range failures in both the first and second samples is three or more, the lot shall be rejected. The test shall be performed as specified in 4.4.1.

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4.3.3.3 Profile and alignment (see dwg. 8839446) - Major defect.- The cartridges shall be gaged 100 percent for profile and alignment. Any cartridge which fails to meet the applicable requirement shall be classed defective and removed from the lot. Equipment shall be in accordance with 4.3.4.

4.3.4 Inspection equipment.-For the performance of the examinations and tests specified in 4.3, when applicable, the contractor shall have available, and utilize correctly, all necessary gages, measuring, and testing equipment (see 6.3). When inspecting the contractor's inspection equipment, the Government inspector will determine that the contractor has available, and utilizes correctly, gages, measuring and testing equipment of the required accuracy and precision and, that the instruments are of the proper type and range to make measurements of the required accuracy. The contractor will have available a set of master gages, standards, and appropriate instruments, for regularly scheduled calibration of his inspection equipment. Records of such regularly scheduled calibration will be maintained by the contractor and made available for review by the Government. The calibration of gages, standards, and instruments will be checked periodically by authorized Government personnel.

4.4 Test methods and procedures.

4.4.1 Proving ground tests.-Twenty of the sample cartridges shall be fired with an ignition cartridge and one propellent increment, and the remaining twenty cartridges shall be fired with an ignition cartridge and eight propellent increments. The cartridges shall be fired from a mortar for which the cartridge is standard. The elevation of the mortar shall be such that the maximum range is obtained for the phase being tested. The impact range shall be recorded for each round fired. In addition, velocity and pressure shall be obtained and recorded, for information only.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.-The cartridge shall be preserved and packaged in accordance with dwg. 7549031.

5.2 Packing.

5.2.1 Level A.-The containers shall be packed in accordance with dwg. 7549032.

5.3 Data cards.-Data card information shall be as specified in Specification MIL-C-2550.

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6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number and date of this specification.

6.2 Cost of check test.-The contracting officer shall arrange for the contractor to be reimbursed for the expense incurred in the performance of the check tests. The test shall be conducted at Government expense without cost to the contractor who loaded the primers or to the contractor assembling the cartridge, and shall not constitute a basis for rejection against either contractor, except where deterioration has occurred as a direct result of carelessness in handling, storage etc., permitted while the primer lot was under the jurisdiction of either contractor.

6.3 Inspection equipment.-Inspection equipment designs were not available at the time this specification was prepared. Prior to procurement, the procuring agency should contact Picatinny Arsenal, Dover, New Jersey, ATTN: ORDBB-NR, to determine availability of inspection equipment designs.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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